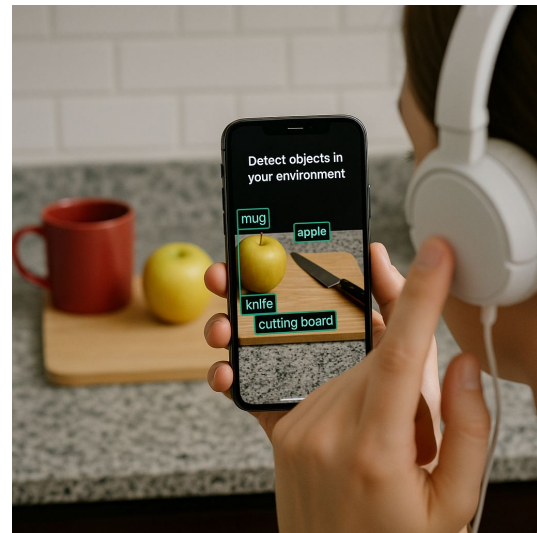


Object Detection

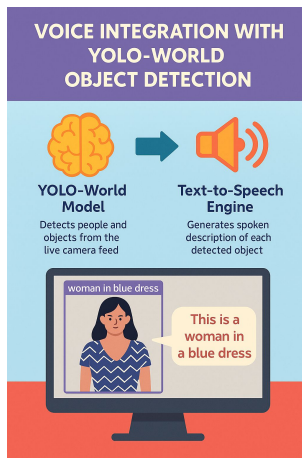
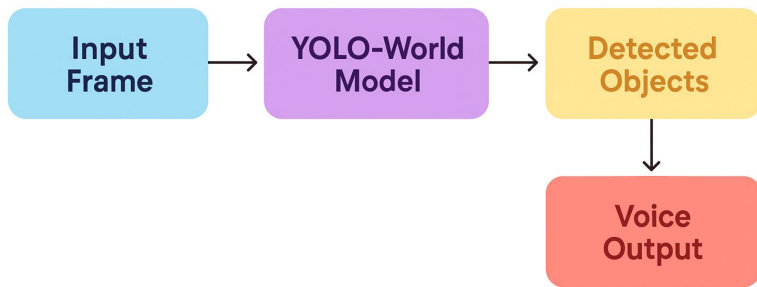
Voice Integration



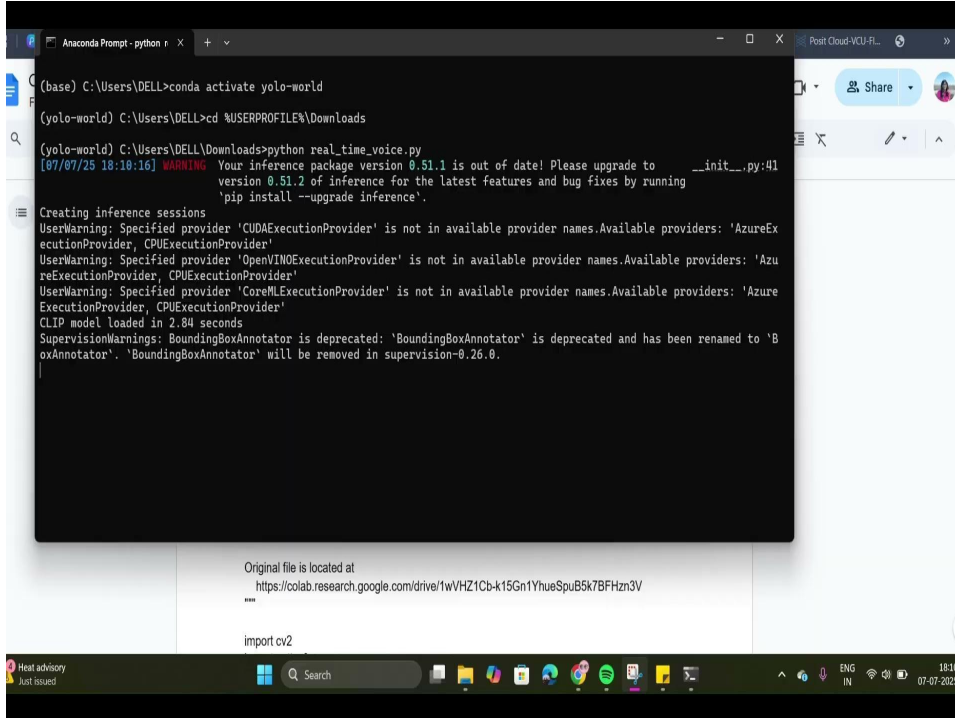
Voice-Integrated Real-Time Object Detection using YOLO-World

To enhance accessibility and interactivity in real-time computer vision systems, we had integrated **object detection with speech output** using **YOLO-World** and **Text-to-Speech (TTS)** technology.

This integration allows the system to not only recognize objects via webcam but also **speak aloud what it sees**, making it useful for visually impaired users, robotics applications, and smart home assistants.



Real Time Recorded Video LIVE



```
(base) C:\Users\DELL>conda activate yolo-world
(yolo-world) C:\Users\DELL>cd %USERPROFILE%\Downloads
(yolo-world) C:\Users\DELL>python real_time_voice.py
[07/07/25 18:10:16] WARNING: Your inference package version 0.51.1 is out of date! Please upgrade to
version 0.51.2 of inference for the latest features and bug fixes by running
'pip install --upgrade inference'.
Creating inference sessions
UserWarning: Specified provider 'CUDAExecutionProvider' is not in available provider names.Available providers: 'AzureEx
ecutionProvider, CPUExecutionProvider'
UserWarning: Specified provider 'OpenVINOExecutionProvider' is not in available provider names.Available providers: 'Azu
reExecutionProvider, CPUExecutionProvider'
UserWarning: Specified provider 'CoreMExecutionProvider' is not in available provider names.Available providers: 'Azure
ExecutionProvider, CPUExecutionProvider'
CLIP model loaded in 2.84 seconds
SupervisionWarnings: BoundingBoxAnnotator is deprecated: 'BoundingBoxAnnotator' is deprecated and has been renamed to 'B
oxAnnotator'. 'BoundingBoxAnnotator' will be removed in supervision-0.26.0.
import cv2
```

Original file is located at
<https://colab.research.google.com/drive/1wVHZ1Cb-k15Gn1YhueSpuB5k7BFHzn3V>



Example Output

Object Detected and Spoken Output:

Person - bounding box

🔊 “Person”

Phone - bounding box

🔊 “Phone”

Umbrella - bounding box

🔊 “Umbrella”

Chair - bounding box

🔊 “Chair”

The object detected will only be voiced out once.



System Architecture

Detected object labels are passed through a **speech pipeline**:

- ❖ New/unique objects are spoken aloud.
- ❖ Repetition is minimized using a memory buffer (`spoken_labels` set).
- ❖ pyttsx3 uses the **SAPI5** driver to synthesize natural voice output.

Pyttsx3: Offline TTS engine for speech synthesis on Windows (via SAPI5). **SAPI5** stands for **Speech Application Programming Interface version 5**.

It's a **Microsoft speech API** built into Windows that allows developers to use **Text-to-Speech (TTS)** and **Speech Recognition** features.

```
engine = pyttsx3.init(driverName='sapi5')
```

Anaconda: Managed environment for seamless package management and hardware compatibility.



Features Achieved



Zero-shot Detection: Works with any English label (Using YOLO-World)



Real-Time Speech: Outputs phrases like “chair” or “phone” in natural voice.



Memory-Optimized TTS: Avoids repeating objects already mentioned within a frame loop.



Offline Operation: Fully functional without internet (ideal for embedded or field use cases).



Use Case Scenarios



Assistive Tech

Helps visually impaired users understand their surroundings through spoken feedback (e.g., “This is a chair”).



Retail & Inventory

Real-time item identification and spoken updates during scanning or auditing.



Defense & Surveillance

Alerts personnel by announcing visual detections like “This is a gun,” “This is a person with a backpack,” or “This is a suspicious object” in secure zones or checkpoints.



Robotics

Enables robots to audibly describe objects they handle or navigate around in real time.



Challenges to Overcome

- Lag and slow detections



Future Scope

- Detect person on the **left**, **right**, and **when leaving the frame**
- If a **weapon is detected**, automatically **take a screenshot**
- **Ignore person bounding boxes**, but assign **color labels to other objects** (e.g., **blue** for watch, **green** for phone)

